

DISCRETE MATHEMATICS

PRIME NUMBER

▣ Prime number: A positive integer P greater than 1 is called prime if the only positive factors of P are 1 and P .

Eg. 2, 3, 5, 7 etc.

▣ Composite number: A positive non-prime integer which is greater than 1 is called a composite number. For e.g. 4, 15 etc.

▣ Relative Prime / Co-prime: Two positive integers are called relative prime numbers / co-prime if their greatest common divisor is 1.

For eg.: 3 and 10, 2 and 7 etc.

▣ Mersenne Prime number: A positive prime number of this form $2^p - 1$ is called a Mersenne prime if p be also a prime number.

For eg.: $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$ are Mersenne primes while $2^{11} - 1 = 2047$ is not since $2047 = 23 \times 89$.

▣ Prove that, if n is a composite number, then n has a prime divisor less than or equal to $\sqrt{n}$.

Solⁿ: If n is composite, then it has a factor ' x ' such that $1 < x \leq n$. Hence $n = xy$ where both x and y are positive integers greater than 1.

It is clear that, either $x \leq \sqrt{n}$ or $y \leq \sqrt{n}$. If x and y are both less than $\sqrt{n}$, then we would get $xy < \sqrt{n} \cdot \sqrt{n} = n$, but that is not possible.

Again, if x and y be both greater than $\sqrt{n}$, then we would get $xy > \sqrt{n} \cdot \sqrt{n} = n$ and that is not possible too.

Either x is less than or equal to $\sqrt{n}$ or y .

Hence n has a positive factor not exceeding $\sqrt{n}$.

This factor can be prime, or by the fundamental theorem of Arithmetic, has a prime divisor.

In either case, n has a prime divisor less than or equal to $\sqrt{n}$.

▣ Prime Factorization: Every positive integer $n > 1$ can be written uniquely as the product of primes,

$$\begin{aligned} \text{Eg. : } 12 &= 2 \cdot 2 \cdot 3 \\ &= 2^2 \cdot 3 \end{aligned}$$

Finding Prime Factorization

i) 289,

$$\begin{array}{r} \text{Quotient} \\ 17 \overline{) 289} \\ \underline{17} \end{array}$$

Hence, $289 = 17^2$

ii) 101,

101 is itself a prime number. Hence,
 $101 = 101^1$

DIVISION ALGO.

Division Algorithm: Let a be an integer and d be any positive integer. Then there exists unique integers q and r where $0 \leq r < d$ such that $a = dq + r$.

Here,

$a \rightarrow$ dividend

$d \rightarrow$ divisor

$q \rightarrow$ quotient

$r \rightarrow$ remainder

Find the quotient and remainder when -11 is divided by 3.

$$\begin{array}{r} 3 \overline{) -11} \quad -4 \\ \underline{-12} \\ 1 \end{array}$$

quotient = -4

remainder = 1

Ex Show that if $a|b$ and $b|c$, then $a|c$.

Soln: Since a divides b , there exists an integer (say) p where $b = a \cdot p$.

Again since b divides c , there exists another integer (say) q where $c = b \cdot q$.

We get,

$$c = b \cdot q$$

$$= a \cdot p \cdot q \quad [\because b = a \cdot p]$$

$$\therefore c = a \cdot pq$$

As pq is also an integer, hence it can be said that a divides c i.e. $a|c$ is true.

GCD & LCM

Ex GCD: Let a and b be integers, not both zero. If d be the largest number such that $d|a$ and $d|b$, then d is called the greatest common divisor (gcd) of a and b , denoted by

$$d = \text{gcd}(a, b)$$

Ex LCM: Let a and b be integers. If d be the smallest integer such that $a|d$ and $b|d$, then d is called the least common multiple (lcm) of a and b , denoted by

$$d = \text{lcm}(a, b)$$

Procedure to find $\text{gcd}(a, b)$: Positive integers)
using Euclidean algorithm.

Soln:

$x := a$

$y := b$

while $y \neq 0$

begin

$r := x \bmod y$

$x := y$

$y := r$

end { $\text{gcd}(a, b)$ is x }

Find the gcd of 252 and 198 using E.A.

Soln:

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18 + 0$$

Hence $\text{gcd}(252, 198) = 18$.

Linear combination: If a and b be positive integers, then there exists integers s and t such that $\text{gcd}(a, b) = sa + tb$, i.e. $\text{gcd}(a, b)$ can be expressed as a linear combination with integer coefficients of a and b .

Ex Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198.

Soln:

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18 + 0$$

Here,

$$18 = 54 - 1 \cdot 36$$

$$\text{and } 36 = 198 - 3 \cdot 54$$

$$\text{So, } 18 = 54 - 1 \cdot (198 - 3 \cdot 54)$$

$$= 54 - 1 \cdot 198 + 3 \cdot 54$$

$$\Rightarrow 18 = 4 \cdot 54 - 1 \cdot 198$$

$$\text{Again, } 54 = 252 - 1 \cdot 198$$

$$\text{So, } 18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198$$

$$= 4 \cdot 252 - 4 \cdot 198 - 1 \cdot 198$$

$$\text{Hence, } 18 = 4 \cdot 252 - 5 \cdot 198$$

Find GCD of 111 and 201 using EA

Soln:

$$201 = 1 \cdot 111 + 90$$

$$111 = 1 \cdot 90 + 21$$

$$90 = 4 \cdot 21 + 6$$

$$21 = 3 \cdot 6 + 3$$

$$6 = 2 \cdot 3 + 0$$

Hence, $\text{GCD}(111, 201) = 3$.

MODULAR ARITHMETIC

Linear Congruence: A congruence of the form $ax \equiv b \pmod{m}$ is called a linear congruence where m is a positive integer, a and b are integers, x is a variable.

Find solution to the linear congruences:

1) $3x \equiv 4 \pmod{7}$

Since $\text{GCD}(3, 7) = 1$, the equation has a unique solution.

Applying Euclidean algorithm to find a solution to

$$3x \equiv 1 \pmod{7}, \text{ --- (1)}$$

We get,

$$\text{GCD}(3, 7) \Rightarrow 7 = 2 \cdot 3 + 1$$

$$3 = 1 \cdot 3 + 0$$

$$\therefore 1 = 7 - 2 \cdot 3$$

$$\text{Or, } 1 = -2 \cdot 3 + 7$$

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That means $s = -2$ is a solution to eqⁿ (1).

$$\begin{aligned}\text{Then, } sb &= -2 \times 4 \\ &= -8\end{aligned}$$

$$\begin{aligned}\text{and } sb \bmod m &= -8 \bmod 7 \\ &= 6\end{aligned}$$

$$\begin{array}{r|l} 7 & -8 \\ \hline & -14 \\ \hline & 6 \end{array}$$

i.e. $x = 6$ is the unique solution to $3x \equiv 4 \pmod{7}$.

General solution is $x + mk$ where $k \in \mathbb{Z}$.

So, the solutions to the given eqⁿ are:

6, 13, 20, ... and -1, -8, -15.

$$2) 48x \equiv 28 \pmod{355}$$

Since $\text{gcd}(48, 355) = 1$, the given eqⁿ has a unique solution. Applying Euclidean algorithm to find a solution to

$$48x \equiv 1 \pmod{355} \quad \text{--- (1)}$$

We get,

$$\text{gcd}(48, 355) \Rightarrow$$

$$355 = 7 \cdot 48 + 19$$

$$48 = 2 \cdot 19 + 10$$

$$19 = 1 \cdot 10 + 9$$

$$10 = 1 \cdot 9 + 1$$

$$9 = 1 \cdot 9 + 0$$

$$\text{So } 1 = 10 - 1 \cdot 9$$

$$= (48 - 2 \cdot 19) - 1 \cdot (19 - 1 \cdot 10)$$

$$= 48 - 2 \cdot 19 - 1 \cdot 19 + 1 \cdot 10$$

$$= 48 - 3 \cdot 19 + 1 \cdot 10$$

$$= 48 - 3 \cdot (355 - 7 \cdot 48) + 1 \cdot (48 - 2 \cdot 19)$$

$$= 48 - 3 \cdot 355 + 21 \cdot 48 + 1 \cdot 48 - 2 \cdot 19$$

$$= 23 \cdot 48 - 3 \cdot 355 - 2 \cdot 19$$

$$= 23 \cdot 48 - 3 \cdot 355 - 2 \cdot (355 - 7 \cdot 48)$$

$$= 23 \cdot 48 - 3 \cdot 355 - 2 \cdot 355 + 14 \cdot 48$$

$$\therefore 1 = 37 \cdot 48 - 5 \cdot 355$$

that means, $s = 37$ is a solution to eqⁿ (1)

$$\text{Then } sb = 37 \times 28$$

$$= 1036$$

$$\text{and } x = sb \bmod m$$

$$= 1036 \bmod 355$$

$$= 326$$

i.e. $x = 326$ is the unique solution to

$$48x \equiv 28 \pmod{355}$$

General solution is $x + mk$ where $k \in \mathbb{Z}$.

So, the solutions to the given eqⁿ are:

$$326, 681, 1036, \dots \text{ and } -29, -384, -739.$$

$$3) 8x \equiv 3 \pmod{10}$$

Since $\text{GCD}(8, 10) = 2 \neq 1$, there is no solution to the given eqn.

$$4) 6x \equiv 8 \pmod{14}$$

Since $\text{GCD}(6, 14) = 2 \neq 1$, there is no solution to the given eqn.

$$5) 6x \equiv 1 \pmod{33}$$

Since $\text{GCD}(6, 33) = 3 \neq 1$, there is no solution to the given eqn.

$$6) 7x \equiv 1 \pmod{9}$$

Since $\text{GCD}(7, 9) = 1$, the given eqn has a unique solution. Applying Euclidean algorithm to find a solution to

$$7x \equiv 1 \pmod{9} \quad \text{--- (1)}$$

we get,

$$\begin{aligned} \text{GCD}(7, 9) \Rightarrow \quad 9 &= 1 \cdot 7 + 2 \\ 7 &= 3 \cdot 2 + 1 \\ 2 &= 1 \cdot 2 + 0 \end{aligned}$$

$$\text{So, } 1 = 7 - 3 \cdot 2$$

$$= 7 - 3 \cdot (9 - 1 \cdot 7)$$

$$= 7 - 3 \cdot 9 + 3 \cdot 7$$

$$\therefore = 4 \cdot 7 - 3 \cdot 9$$

That means $x = 4$ is a solution to eqn (1).

Thus, the general solution is $4+9k$ where $k \in \mathbb{Z}$. So, the solutions are: $4, 13, 22, \dots$ and $-5, -14, -23, \dots$

7) $81x \equiv 1 \pmod{256}$

Since $\gcd(81, 256) = 1$, there exists a unique solution to the given eqⁿ. Applying Euclidean algorithm to find the $\gcd(81, 256)$, we get,

$$256 = 3 \cdot 81 + 13$$

$$81 = 6 \cdot 13 + 3$$

$$13 = 4 \cdot 3 + 1$$

$$3 = 3 \cdot 1 + 0$$

Here, $1 = 13 - 4 \cdot 3$

and $3 = 81 - 6 \cdot 13$

$$\therefore 1 = 13 - 4 \cdot (81 - 6 \cdot 13)$$

$$= 13 - 4 \cdot 81 + 24 \cdot 13$$

$$= 256 - 3 \cdot 81 - 4 \cdot 81 + 24 \cdot (256 - 3 \cdot 81)$$

$$[\because 13 = 256 - 3 \cdot 81]$$

$$= 256 - 7 \cdot 81 + 24 \cdot 256 - 72 \cdot 81$$

$$= 25 \cdot 256 - 79 \cdot 81$$

So, $s = -79$ is the unique solution to the given eqⁿ. General eqⁿ is $-79 + 256k$ where $k \in \mathbb{Z}$. So, the solutions are: $-79, 177, 433, \dots$ and $-335, -591, -847, \dots$

Find the inverse of 5 modulo 11.

Soln: Since $\text{gcd}(5, 11) = 1$, there exists an inverse of 5 modulo 11.

$$11 = 2 \cdot 5 + 1$$

$$5 = 1 \cdot 5 + 0$$

$$\therefore 1 = -2 \cdot 5 + 11$$

this shows that -2 is an inverse of 5 modulo 11.

Find the inverse of 4 modulo 9.

Soln: Since $\text{gcd}(4, 9) = 1$, there exists an inverse of 4 modulo 9.

$$9 = 2 \cdot 4 + 1$$

$$\therefore 1 = -2 \cdot 4 + 9$$

This shows that -2 is an inverse of 4 modulo 9.

Chinese Remainder Theorem

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers. The system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $M = m_1 m_2 \dots m_n$.

that is there is a solution x with $0 \leq x < M$.

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Other solutions are congruent modulo M to this solution.

Find solution to the systems :

$$\begin{aligned} 1) \quad x &\equiv 2 \pmod{3} \\ x &\equiv 4 \pmod{7} \\ x &\equiv 6 \pmod{10} \end{aligned}$$

Given,

$$m_1 = 3,$$

$$\text{and, } a_1 = 2,$$

$$m_2 = 7,$$

$$a_2 = 4,$$

$$m_3 = 10.$$

$$a_3 = 6.$$

$$\text{So, } M = m_1 \times m_2 \times m_3$$

$$= 3 \times 7 \times 10$$

$$= 210$$

$$\text{and } M_1 = \frac{M}{m_1} = \frac{210}{3} = 70$$

$$M_2 = \frac{M}{m_2} = \frac{210}{7} = 30$$

$$M_3 = \frac{M}{m_3} = \frac{210}{10} = 21$$

Now, we have to seek the solution to the eqn :

$$70x \equiv 1 \pmod{3}$$

$$30x \equiv 1 \pmod{7}$$

$$21x \equiv 1 \pmod{10}$$

Reducing $70 \bmod 3$, $30 \bmod 7$, $21 \bmod 10$,
we have,

$$x \equiv 1 \pmod{3}$$

$$2x \equiv 1 \pmod{7}$$

$$x \equiv 1 \pmod{10}$$

Solution to these eqⁿ,

$$s_1 = 1$$

$$s_2 = 2$$

$$s_3 = 1$$

$$\therefore x = M_1 s_1 a_1 + M_2 s_2 a_2 + M_3 s_3 a_3$$

$$= (70 \times 1 \times 2) + (30 \times 2 \times 4) + (21 \times 1 \times 6)$$

$$= 506$$

Dividing this solution by $M = 210$, we obtain the
unique solution to the given system i.e.,

$$x = x \bmod M$$

$$= 506 \bmod 210$$

$$= 86$$

(Ans).

▣ Pseudorandom Numbers : Numbers generated by systematic methods are not truly random. They are called Pseudorandom numbers.

▣ Generating Pseudorandom Numbers :

1) Find 5 number by choosing $m=17, a=5, c=2, x_0=3$.

Soln. The formula to generate pseudorandom numbers

$$\text{is, } x_{n+1} = (ax_n + c) \bmod m$$

Here, $m=17,$

$$a=5,$$

$$c=2,$$

$$x_0=3$$

$$\therefore x_1 = x_{0+1} = (ax_0 + c) \bmod m$$

$$= (5 \times 3 + 2) \bmod 17$$

$$= 17 \bmod 17$$

$$= 0$$

$$x_2 = x_{1+1} = (ax_1 + c) \bmod m$$

$$= (5 \times 0 + 2) \bmod 17$$

$$= 2 \bmod 17$$

$$= 2$$

$$x_3 = x_{2+1} = (ax_2 + c) \bmod m$$

$$= (5 \times 2 + 2) \bmod 17$$

$$= 12 \bmod 17$$

$$= 12$$

$$\begin{aligned}
 x_4 = x_{3+1} &= (ax_3 + c) \bmod m \\
 &= (5 \times 12 + 2) \bmod 17 \\
 &= 62 \bmod 17 \\
 &= 11
 \end{aligned}$$

$$\begin{aligned}
 x_5 = x_{4+1} &= (ax_4 + c) \bmod m \\
 &= (5 \times 11 + 2) \bmod 17 \\
 &= 57 \bmod 17 \\
 &= 6
 \end{aligned}$$

So, the sequence is 0, 2, 12, 11, 6

~~sequence~~

So, the sequence is 3, 0, 2, 12, 11, ...

III Cryptology : Cryptology is the study of secret messages. One of the earliest known uses of cryptology was by Julius Caesar. He made messages secret by shifting each letter three letters forward in the alphabet. This method can be represented by the function:

$$f(p) = (p+3) \bmod 26 ; \text{ where } 0 \leq p \leq 25$$

IV Fermat's Little theorem : If p be a prime number and a is an integer not divisible by p , then,

$$a^{p-1} \equiv 1 \pmod{p}$$

Furthermore, for every integer a , we have

$$a^p \equiv a \pmod{p}$$

IV Principles of Mathematical Induction

1st P.:

Basic step : We verify that $P(1)$ is true.

Inductive step : $P(n+1)$ is true whenever $P(n)$ is true.
Then P is true for every integers $n \geq 1$.

2nd P.:

Let P be a proposition which is defined on integers $n \geq 1$ such that

1) $P(1)$ is true

2) $P(n+1)$ is true whenever $P(k)$ is true for all $1 \leq k \leq n$.

Then P is true for every integer $n \geq 1$.

Q1 Recursive Functions

$$2) f(0) = 3, f(n+1) = 2f(n) + 3.$$

Find $f(1), f(2), f(3), f(4)$

Solⁿ:

Given, $f(n+1) = 2f(n) + 3$

$$\text{and } f(0) = 3$$

$$\therefore f(1) = f(0+1) = 2 \times f(0) + 3$$

$$= (2 \times 3) + 3$$

$$= 9$$

$$f(2) = f(1+1) = 2 \times f(1) + 3$$

$$= (2 \times 9) + 3$$

$$= 21$$

$$f(3) = f(2+1) = 2f(2) + 3$$

$$= (2 \times 21) + 3$$

$$= 45$$

$$f(4) = f(3+1) = 2f(3) + 3$$

$$= (2 \times 45) + 3$$

$$= 93$$

2) Find $f(1)$, $f(2)$, $f(3)$, $f(4)$ while $f(0) = 1$.

i) $f(n+1) = f(n) + 2$

ii) $f(n+1) = 3f(n)$

iii) $f(n+1) = f(n)^2 + f(n) + 1$

Solⁿ:

Given,

$$f(0) = 1$$

i) $f(n+1) = f(n) + 2$

$$\begin{aligned}\therefore f(1) &= f(0+1) = f(0) + 2 \\ &= 1 + 2 \\ &= 3\end{aligned}$$

$$f(2) = f(1+1) = f(1) + 2 = 3 + 2 = 5$$

$$f(3) = f(2+1) = f(2) + 2 = 5 + 2 = 7$$

$$f(4) = f(3+1) = f(3) + 2 = 7 + 2 = 9$$

ii) $f(n+1) = 3f(n)$

$$\therefore f(1) = f(0+1) = 3f(0) = 3 \times 1 = 3$$

$$f(2) = f(1+1) = 3f(1) = 3 \times 3 = 9$$

$$f(3) = f(2+1) = 3f(2) = 3 \times 9 = 27$$

$$f(4) = f(3+1) = 3f(3) = 3 \times 27 = 81$$

The Direct Method:

$$\text{iii) } f(n+1) = f(n)^2 + f(n) + 1$$

$$\begin{aligned} \therefore f(1) &= f(0+1) = f(0)^2 + f(0) + 1 \\ &= 1^2 + 1 + 1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} f(2) &= f(1+1) = f(1)^2 + f(1) + 1 \\ &= 3^2 + 3 + 1 \\ &= 13 \end{aligned}$$

$$\begin{aligned} f(3) &= f(2+1) = f(2)^2 + f(2) + 1 \\ &= 13^2 + 13 + 1 \\ &= 183 \end{aligned}$$

$$\begin{aligned} f(4) &= f(3+1) = f(3)^2 + f(3) + 1 \\ &= 183^2 + 183 + 1 \\ &= 33673 \end{aligned}$$

Direct Proof: In mathematics and logic, a direct proof is a way of showing the truth or falsehood of a given statement by a straightforward combination of established facts, usually axioms, existing formulas and theorems without making any further assumptions.

Ith Indirect Proof: An indirect proof may begin with the premises, continue with a sequence of deductions, and end with the conclusion.

Ith Proof by Contradiction: In logic proof by contradiction is a form of proof and more specifically a form of indirect proof that establishes the truth or validity of a proposition. It stands by assuming that the opposite proposition is true and shows that such an assumption leads to a contradiction.

COUNTING

Inclusion & Exclusion

When two tasks can be done at the same time, we cannot use the sum rule to count the number of ways to do one of the two tasks. Adding the number of ways to do each task leads to an over count, since the easy to do both tasks are counted twice. To correctly count the number of ways to do one of the two task, we add the number of ways to do each of the two tasks and then subtract the number of ways to do both tasks. This technique is called principle of inclusion and exclusion.

Eg.: To find, how many positive integers from 1 to 100 are either multiples of 2 or multiples of 3.

There are $\lfloor 100/2 \rfloor = 50$ numbers between 1 & 100 those are multiples of ~~100~~ 2.

Again there are $\lfloor 100/3 \rfloor = 33$ numbers between 1 & 100 those are multiples of 3.

And there are $\lfloor 100/(2 \times 3) \rfloor = \lfloor 100/6 \rfloor = 16$ numbers between 1 and 100 those are multiples of both 2 and 3.

Thus number of positive integers from 1 to 100 whose are either multiples of 2 or multiples of 3 is

$$50 + 33 - 16 = 67.$$

Q How many integers from 1 to 1000 are either multiples of 3 or multiples of 5?

Solⁿ: There are $\lfloor 1000/3 \rfloor = 333$ numbers between 1 & 1000 those are multiples of 3.

There are $\lfloor 1000/5 \rfloor = 200$ numbers between 1 & 1000 those are multiples of 5.

There are $\lfloor 1000/(3 \times 5) \rfloor = \lfloor 1000/15 \rfloor = 66$ numbers between 1 and 1000 those are multiples of both 3 and 5.

Hence, there are $(333 + 200 - 66) = 467$ integers from 1 to 1000 whose are either multiples of 3 or multiples of 5.

Q How many bit strings of length seven either start with the two bits 11 or end with a 0.

Solⁿ: We can construct $2^5 = 32$ bit strings starting with the two bits 11.

We can construct $2^6 = 64$ bit strings ending with a 0.

We can construct $2^4 = 16$ bit strings starting with the two bits 11 and ending with a 0.

So, we can construct $(32 + 64 - 16) = 80$ bit strings either starting with the two bits 11 or ending with a 0.

Sum Rule: If a ~~task~~ first task can be done in n_1 ways and a second task in n_2 ways and if these tasks can't be done at the same time, then there are $n_1 + n_2$ ways to do either task.

Product Rule: Let us consider that a procedure can be broken down into two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 \cdot n_2$ ways to do the procedure.

Pigeonhole Principle: If ~~at~~ $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

Proof: Let us assume that none of the k boxes contains more than one object. Then the total number of objects would be at most k . This is a contradiction, since there are at least $k+1$ objects.

Generalized Form: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Find the number of minimum students required in a DM class to be sure that at least six will receive the same grade, if there are five possible grades, namely A, B, C, D, F.

Solⁿ: The ~~min~~ minimum number of students needed to guarantee that at least six students receive the same grade is the smallest integer N such that $\lceil N/5 \rceil = 6$.
The smallest such integer $N = 5 \cdot 5 + 1 = 26$.

Am.

How many ten letter words can be formed using the letters of the word "BANGLADESH"?

Solⁿ: There are 2 A's.

$$\text{The number of total ten-letter words} = \frac{10!}{2!}$$

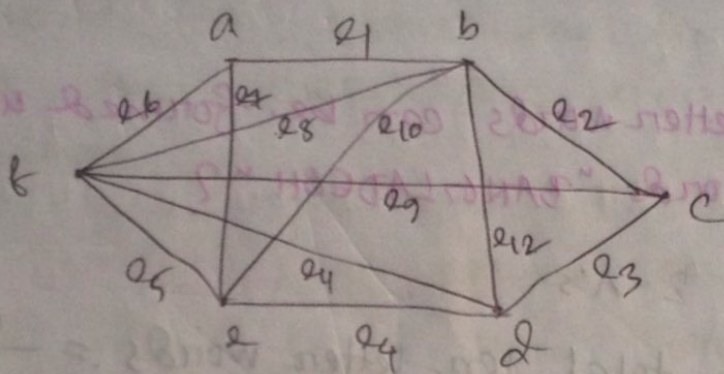
$$= 1814400$$

Am.

GRAPH

Graph

A graph $G = (V, E)$ consists of V , a non-empty set of vertices and E , a set of edges. Each edge has either one or two vertices associated with it, called endpoints. An edge is said to connect its endpoints.



Adjacency

	a	b	c	d	e	f
a	0	1	0	0	1	1
b	1	0	1	1	1	1
c	0	1	0	1	0	1
d	0	1	1	0	1	1
e	1	1	0	1	0	1
f	1	1	1	1	1	0

Incident Matrix

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8	e_9	e_{10}	e_{11}	e_{12}
a	1	0	0	0	0	1	1	0	0	0	0	0
b	1	1	0	0	0	0	0	1	0	1	0	0
c	0	1	1	0	0	0	0	0	1	0	0	0
d	0	0	1	1	0	0	0	0	0	0	1	1
e	0	0	0	1	1	0	1	0	0	1	0	0
f	0	0	0	0	1	1	0	1	1	0	1	0

#

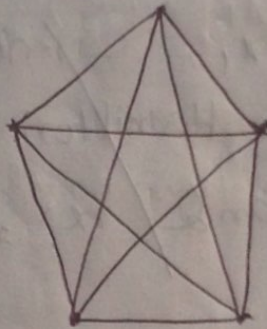
Hamilton Path: A path $x_0, x_1, \dots, x_{n-1}, x_n$ in a graph $G = (V, E)$ is called Hamilton path if $V = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ and $x_i \neq x_j$ for $0 \leq i < j \leq n$.

A simple path in a graph G that passes through every vertex exactly once is called a Hamilton path, and a simple circuit in a graph G that passes through every vertex exactly once is called a Hamilton circuit.

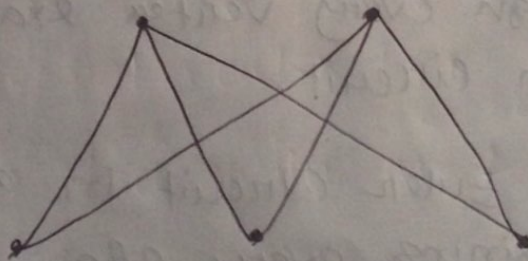
Eulerian Circuit: An Euler circuit in a graph is a simple circuit containing every edge of G .
An Euler Path in G is a simple path containing every edge of G .

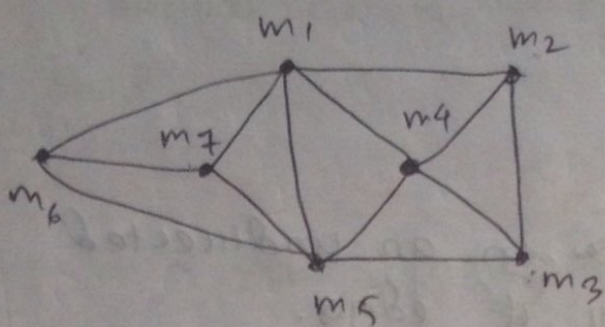
Bipartite Graph: A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 so that no edge in G connects either two vertices in V_1 or two vertices in V_2 . When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G .

#5

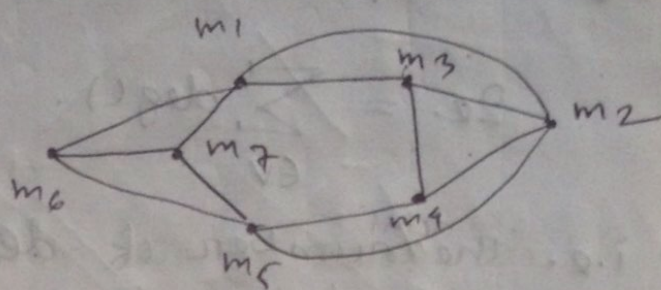


$K_{2,3}$





G1



H

G2

	m ₁	m ₂	m ₃	m ₄	m ₅	m ₆	m ₇
m ₁	0	1	0	1	1	1	1
m ₂	1	0	1	1	0	0	0
m ₃	0	1	0	1	1	0	0
m ₄	1	1	1	0	1	0	0
m ₅	1	0	1	1	0	1	1
m ₆	1	0	0	0	1	0	1
m ₇	1	0	0	0	1	1	0

Handshaking Theorem:

Let $G = (V, E)$ be an undirected graph with e edges. Then,

$$2e = \sum_{v \in V} \deg(v).$$

i.e. the ^{total} number of degree in an undirected graph is twice of number of edges.

Tree: A tree is a connected undirected graph with no simple circuit.

Rooted Tree: A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

⊛ A rooted tree is called an m-ary tree if every internal vertex has no more than m children.

The tree is called a full m-ary tree if every internal vertex has exactly m children.

An m-ary tree with $m=2$ is called a binary tree.

Internal Node: Nodes that have children are called internal nodes.

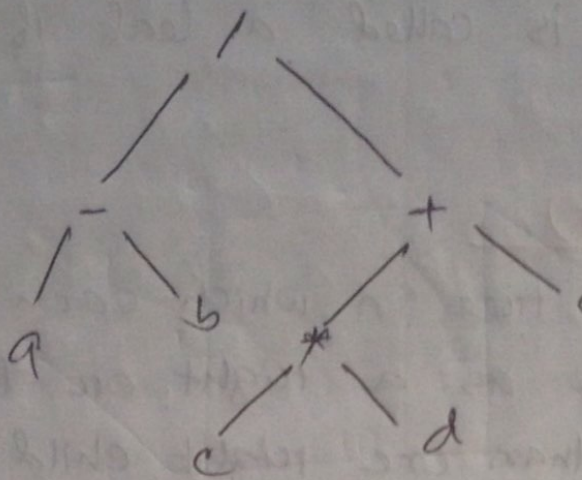
Leaf: A node of a tree is called a leaf if it has no child.

Binary Search Tree (BST)

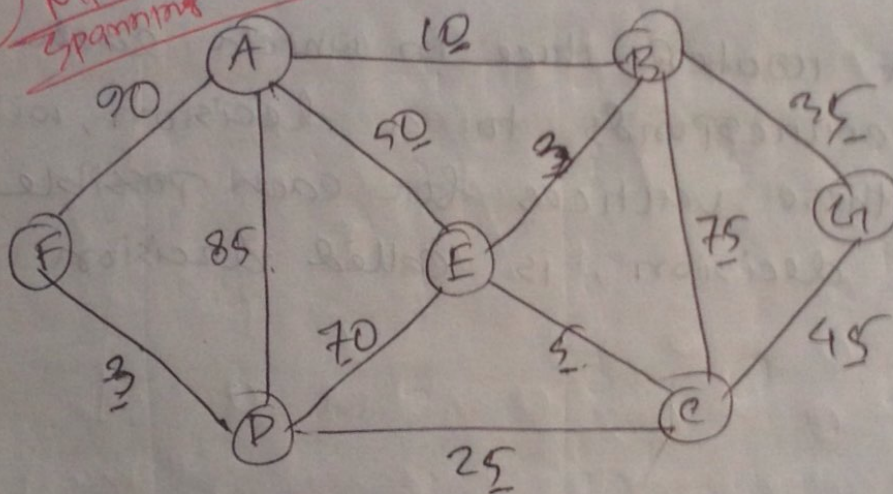
BST is a ~~tree~~ binary tree in which each child of a vertex is designated as a right or left child, no vertex has more than one right child or left child, and each vertex is labeled with a key, which is one of the items.

Decision Tree: A rooted tree in which each internal vertex corresponds to a decision, with a subtree at these vertices for each possible outcome of the decision, is called decision tree.

$$⑧ (a-b) / ((c * d) + e)$$

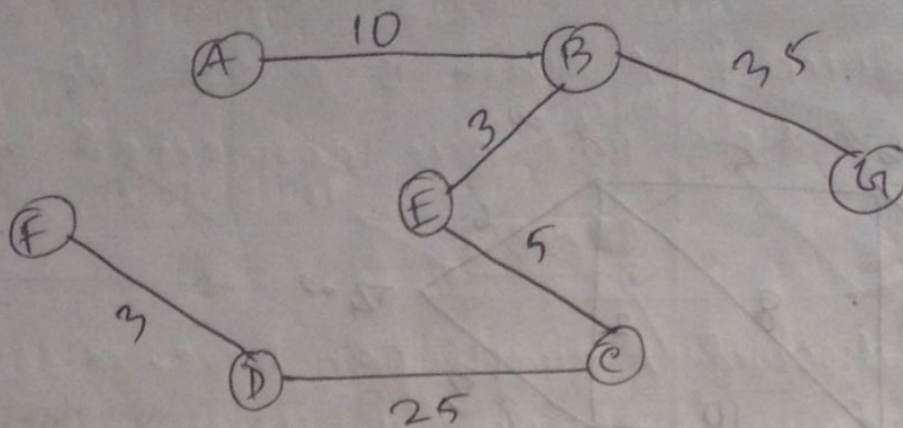


⑧ Min Cost Spanning Tree



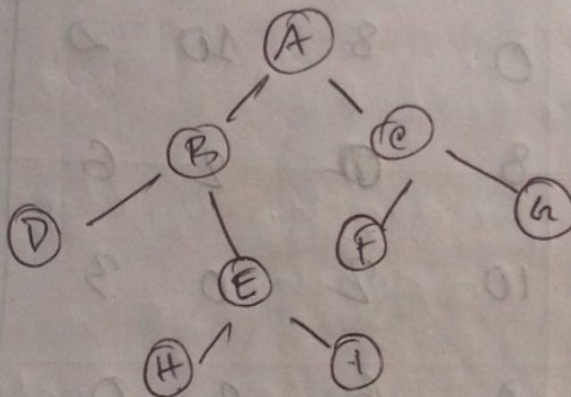
Sorting in ascending form ~~we~~ we get the cost of the edges :

3, 3, 5, 10, 25, 35, 45, 50, 70, 75, 85, 90



$$\text{Minimum cost} = 3 + 3 + 5 + 10 + 25 + 35 = 81$$

Tree Traversal



Pre Order : (Root - Left - Right)

A → B → D → E → H → I → C → F → G

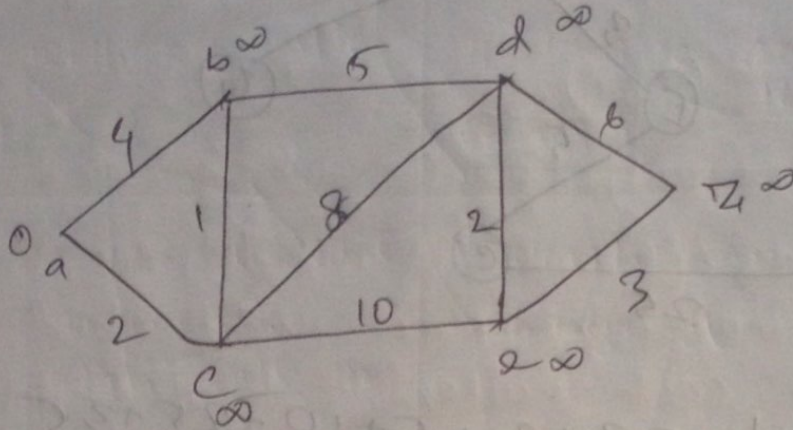
In Order : (Left - Root - Right)

D → B → H → E → I → F → C → G

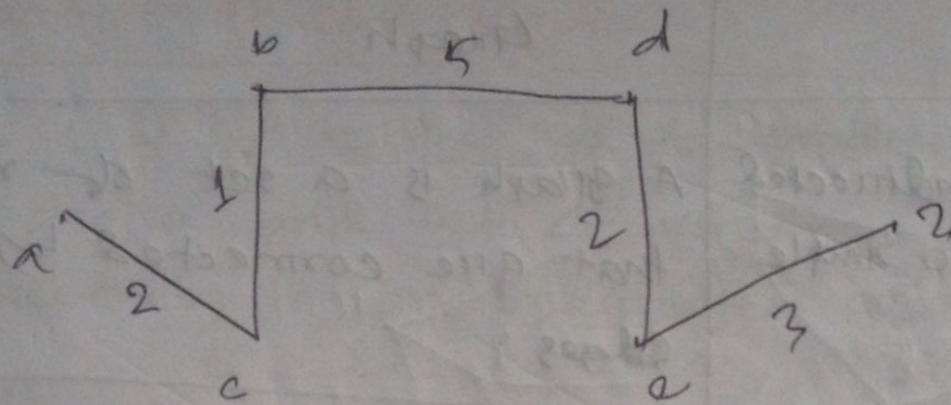
Post Order (Left-Right-Root)

$D \rightarrow H \rightarrow I \rightarrow E \rightarrow B \rightarrow F \rightarrow G \rightarrow C \rightarrow A$

① Dijkstra

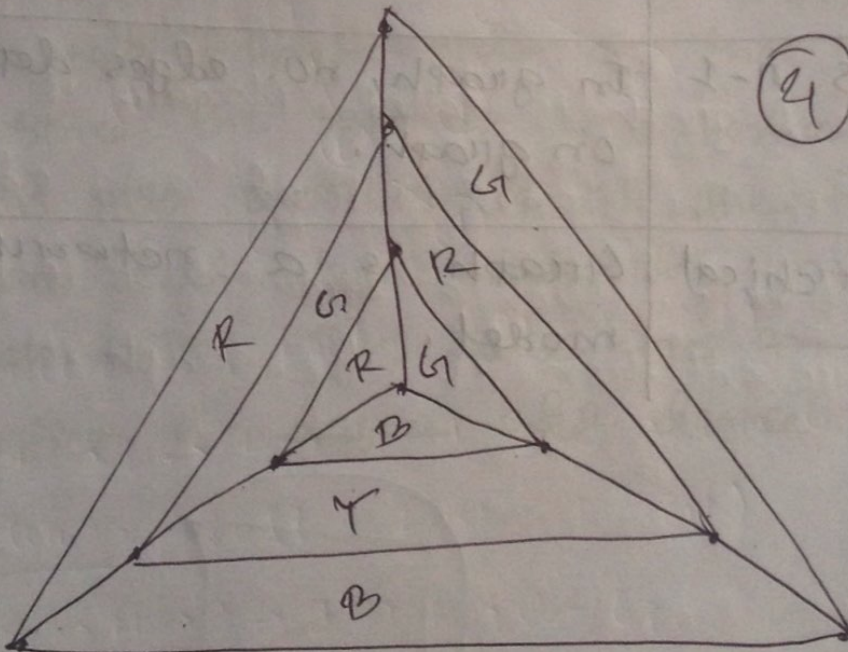


	a	b	c	d	e	g
a	0	4	2	∞	∞	∞
b	4	0	1	5	∞	∞
c	2	1	0	8	10	∞
d	∞	5	8	0	2	6
e	∞	∞	10	2	0	3
g	∞	∞	∞	6	3	0



Shortest Path is $a \rightarrow c \rightarrow b \rightarrow d \rightarrow e \rightarrow f$ with length
 $2 + 1 + 5 + 2 + 3 = 13$.

④ Chromatic Number: the chromatic number of a graph is the least number of colors needed for coloring of ~~this~~ the graph.



④

Diff betⁿ Tree & Graph

Tree	Graph
A connected undirected graph with no simple circuit	A graph is a set of nodes that are connected by edges.
A tree can be described as a specialized case of graph with no self loops & circuits	A graph can have a number of loops
In tree, there is exactly one root node and every child have only one parent.	In graph, there is no such concept of root node.
Tree always has $n-1$ edges.	In graph, no. edges depends on graph.
Tree is a hierarchical model	Graph is a network model.

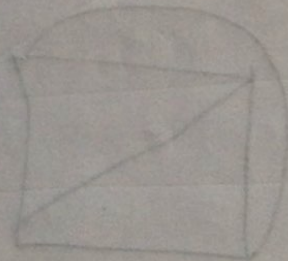
④ An undirected graph has an even number of vertices of odd degree.

Let V_1 & V_2 be the sets of vertices of even degree and set of vertices of odd degree respectively in an undirected graph $G=(V, E)$.

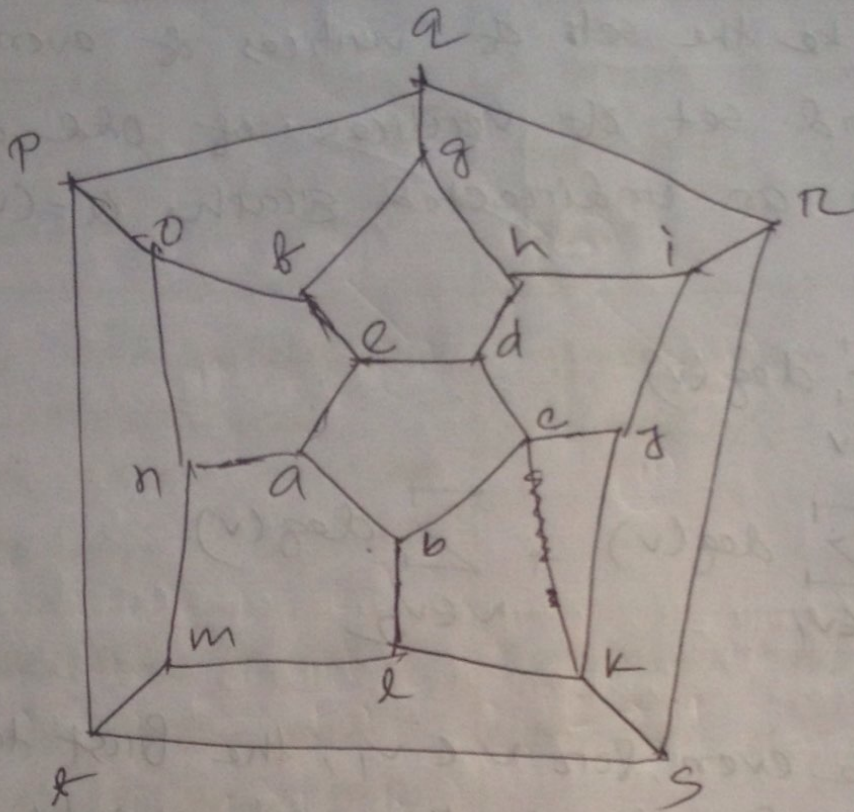
Then,

$$\begin{aligned} 2e &= \sum_{v \in V} \deg(v) \\ &= \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v) \end{aligned}$$

Since $\deg(v)$ is even for $v \in V_1$, the first term in the right hand side of the last quantity is even. Furthermore, the sum of the two terms on the right hand side of the last quantity is even. Since the sum is $2e$, hence the second term is also even. Since all the terms in this sum are odd, there must be an even number of such terms. Thus there are an even number of vertices of odd degree.



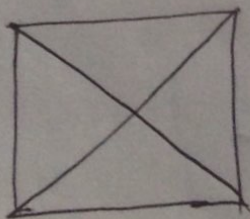
Hamilton



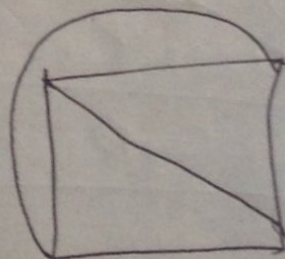
a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t

Planar Graph: A graph is called planar if it can be drawn in the plane without any edges crossing each other. Such a drawing is called a planar representation of the graph.

Ex:

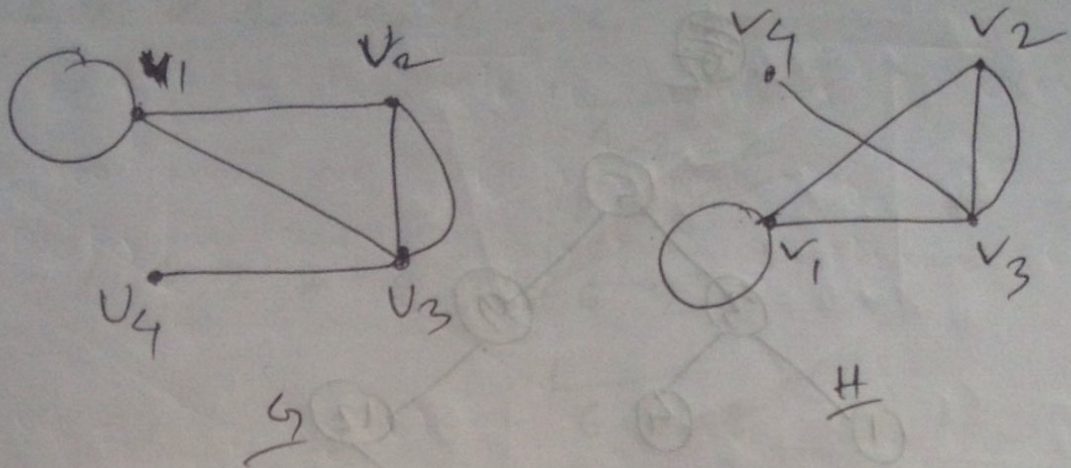


K_4



Planar Repre. of K_4

Isomorphic :



Both graphs have 4 vertices.

$f(v_1) = v_1 = 4$ vertices

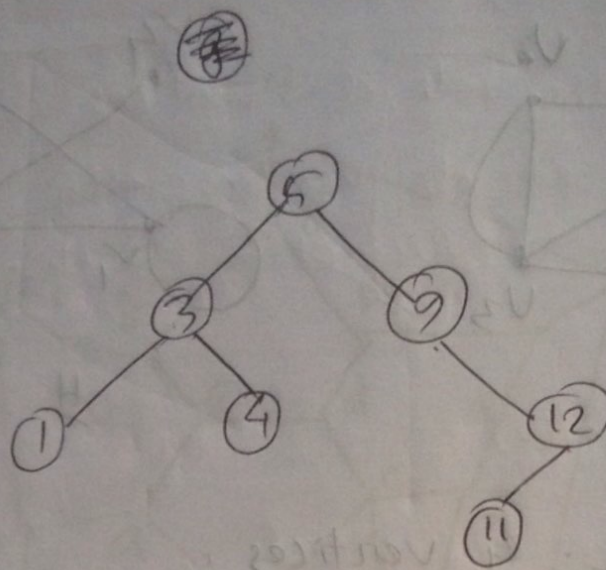
$f(v_2) = v_2 = 3$ vertices

$f(v_3) = v_3 = 4$ vertices

$f(v_4) = v_4 = 1$ vertices

Hence, they are isomorphic.

Spanning Tree : Let G be a simple graph. A spanning tree of G is a subgraph of G that is a tree containing every vertex of G .



Pre: (Root - L - R)

5 → 3 → 1 → 4 → 9 → 12 → 11

In: (Left - Root - R)

1 → 3 → 4 → 5 → 9 → 12 → 11

Post: (L - Right - Root)

1 → 4 → 3 → 11 → 12 → 9 → 5

① Prefix

$$\textcircled{1} + - / 6 2 * 2 3 + 6 - 4 2$$

$$\begin{array}{r} \text{---} \\ 4 - 2 = 2 \end{array}$$

$$= + - / 6 2 * 2 3 + 6 2$$

$$\begin{array}{r} \text{---} \\ 6 + 2 = 8 \end{array}$$

$$= + - / 6 2 * 2 3 8$$

$$\begin{array}{r} \text{---} \\ 2 * 3 = 6 \end{array}$$

$$= + - / 6 2 6 8$$

$$\begin{array}{r} \text{---} \\ 6 / 2 = 3 \end{array}$$

$$= + - 3 6 8$$

$$\begin{array}{r} \text{---} \\ 6 - 3 = 3 \end{array}$$

$$= + 3 8$$

$$\begin{array}{r} \text{---} \\ 8 + 3 = 11 \end{array}$$

$$= 11$$

②

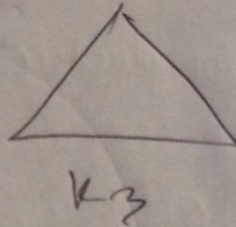
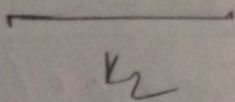
$$\begin{array}{c} \text{---} * 2 / 8 \ 4 \ 3 \\ \quad \quad \quad \underline{\hspace{2cm}} \\ \quad \quad \quad 8 / 4 = 2 \end{array}$$

$$\begin{array}{c} \text{---} * 2 \ 2 \ 3 \\ \quad \quad \quad \underline{\hspace{2cm}} \\ \quad \quad \quad 2 * 2 = 4 \end{array}$$

$$\begin{array}{c} \text{---} 4 \ 3 \\ \quad \quad \quad \underline{\hspace{2cm}} \\ \quad \quad \quad 4 - 3 = 1 \end{array}$$

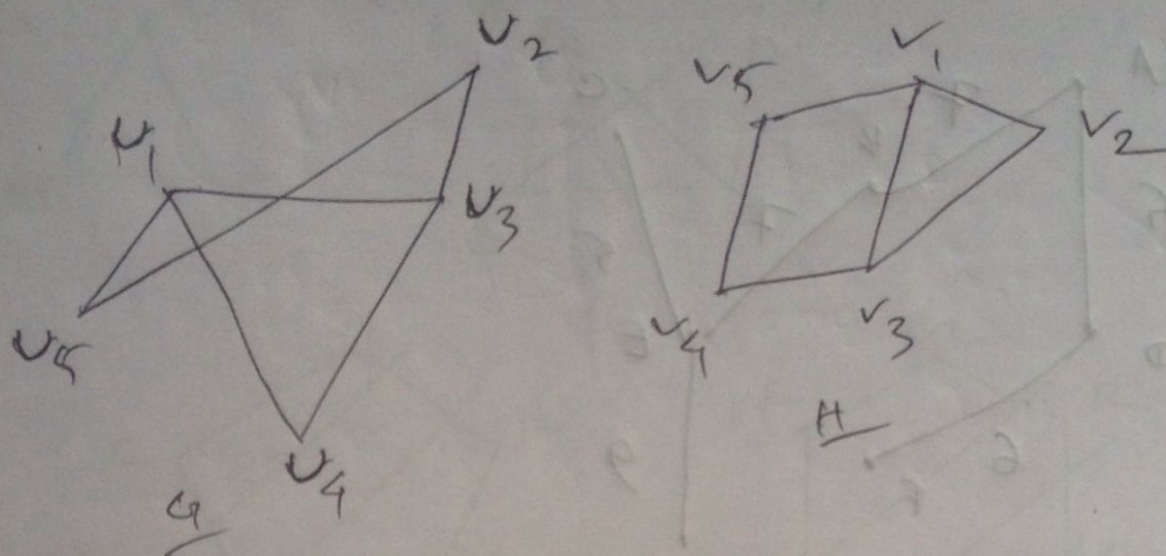
$$= 1$$

⑧ Complete Graph: A complete graph of n vertices denoted by K_n is a simple graph that contains only edges between each pair of distinct vertex.



⑨ Isomorphism: The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there is a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 iff $f(a)$ and $f(b)$ are adjacent in G_2 for all a and b in V_1 . Such

a function f is called an isomorphism.



Here $U_1 = V_1 = 3$, $U_2 = V_2 = 2$, $U_3 = V_3 = 3$,
 $U_4 = V_4 = 2$, $U_5 = V_5 = 2$.

So, G & H are isomorphic.

Pre Order : (Root - L - R)

$a \rightarrow b \rightarrow i \rightarrow j \rightarrow c \rightarrow d \rightarrow f \rightarrow k \rightarrow g \rightarrow h \rightarrow l \rightarrow m$

In order : (L - Root - R)

$i \rightarrow c \rightarrow j \rightarrow b \rightarrow a \rightarrow l \rightarrow k \rightarrow f \rightarrow d \rightarrow g \rightarrow h \rightarrow m$

Post L - Right - Root :

$i \rightarrow j \rightarrow d \rightarrow b \rightarrow c \rightarrow k \rightarrow f \rightarrow g \rightarrow l \rightarrow m \rightarrow h \rightarrow a$

